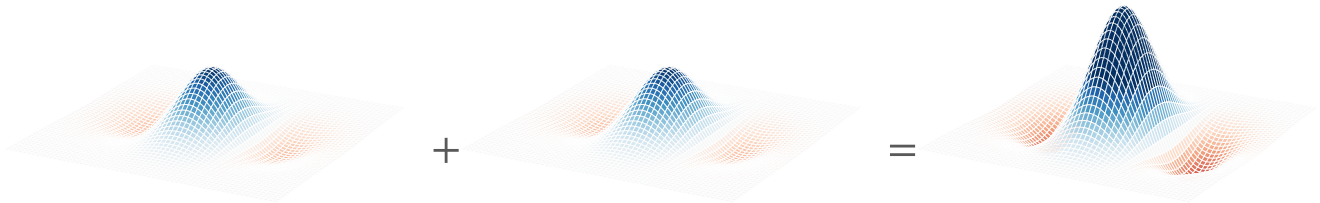
Constructive: $c_2 = +c_1$

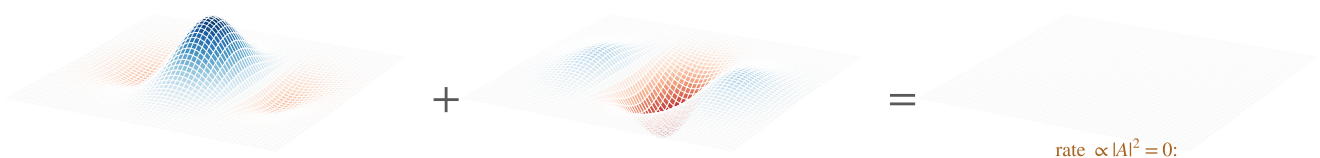
$A_1 = \kappa c_1 u(x, y)$ $A_2 = \kappa c_2 u(x, y)$ $A = A_1 + A_2$



rate $\propto |A|^2 = 4|A_1|^2$

Destructive: $c_2 = -c_1$

$A_1 = \kappa c_1 u(x, y)$ $A_2 = \kappa c_2 u(x, y)$ $A = A_1 + A_2$



rate $\propto |A|^2 = 0$:
dark to this channel;
excitation remains in the
orthogonal mode combination

Signed detection amplitudes for one excitation (blue > 0, red < 0); not field energy and not photon energy